

IMPLICIT MULTI-RESOLUTION DUAL-GRAPH VERIFICATION FOR DECISION TREE ENSEMBLES

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ABSTRACT

Formal verification of decision tree ensembles—such as gradient boosted decision trees (GBDTs) and random forests—is critical for certifying safety, fairness, and adversarial robustness in high-stakes tabular applications. Existing exact verifiers, however, face a fundamental dilemma: constraint-based solvers (MILP/SMT) incur prohibitive per-query latencies, whereas spatial frameworks explicitly materialize the output-configuration (OC) space, scaling as $\mathcal{O}(|\mathcal{O}| \cdot F)$ in memory—causing catastrophic exhaustion on large models—while failing to support dynamic model updates. We present **Implicit Multi-Resolution Dual-Graph Verification (IMD-GV)**, an engine that lazily constructs dual-simplex adjacency edges over a multi-resolution macro-cell hierarchy. IMD-GV navigates this implicit state space via a best-first branch-and-bound walk that synergistically combines spatial distance lower bounds with per-tree achievable-class output-feasibility bounds. An adaptive hybrid selector routes queries using analytical bounds, and localized macro-cell refinement enables millisecond incremental updates under structural edits (pruning, appending, and re-splitting trees). Across 16 benchmark datasets and 1,065 Pareto-compressed ensembles, IMD-GV attains exact agreement with ground truth, deterministic resident box-memory reductions up to $418,316\times$ (median $95\times$), a $12.1\times$ median per-query speedup over in-RAM OC-Tree indexes (100–400 $\times$ faster than MILP), and a $20,918\times$ median speedup for structural model edits, generalizing across L_∞ , L_1 , L_2 , and hybrid metrics without reformulation.

1 INTRODUCTION

Tree ensembles, such as gradient boosted decision trees (GBDTs) and random forests, are among the most effective and widely deployed models for tabular decision-making in critical domains, including medical diagnostics, credit scoring, and automated policy control. Because these models directly inform high-stakes decisions, establishing provable guarantees regarding their safety, fairness, and adversarial robustness is a prerequisite for real-world deployment (Urban & Min’e, 2021; Liu et al., 2019). Unlike continuous deep neural networks, decision tree ensembles partition the input space into non-convex, piecewise-constant decision regions through orthogonal split hyperplanes. Consequently, verifying exact properties over tree ensembles introduces distinct combinatorial challenges (Kantchelian et al., 2015; Chen et al., 2019; Devos et al., 2020).

Existing exact verification frameworks for tree-based models generally follow one of two paradigms, each burdened by a severe computational or spatial trade-off:

1. **Constraint-based optimization solvers (MILP & SMT).** These methods encode the ensemble and the verification target into Mixed-Integer Linear Programs (Kantchelian et al., 2015) or Satisfiability Modulo Theories formulas (Einziger et al., 2019). While exact, generic solvers treat the model as a static constraint set, incurring prohibitive per-query latencies (often 100–700 $\times$ slower than spatial search). Adapting the formulation to custom metrics (e.g., hybrid L_∞/L_1 budgets) requires substantial mathematical re-derivation.
2. **Explicit OC-space enumeration & indexing.** To avoid generic-solver overhead, recent work frames verification as a unified spatial search over the output-configuration (OC)

space (Devos et al., 2020; Sato et al., 2019). An OC is a convex hyper-rectangle formed by intersecting leaf boxes across all trees, within which the raw prediction is constant. Existing frameworks pre-materialize all valid OCs into spatial structures such as R^* -trees (Beckmann et al., 1990), Rootbox, or OC-Trees (Devos et al., 2020). Storing the explicit OC-space requires $\mathcal{O}(|\mathcal{O}| \cdot F)$ memory ($|\mathcal{O}|$ = number of valid OCs, F = feature dimensionality), causing memory exhaustion on hundred-million-cell spaces. Moreover, any structural edit invalidates the index, forcing costly re-enumeration.

To overcome the dual bottlenecks of explicit memory allocation and static constraint solving, we introduce the **Implicit Multi-Resolution Dual-Graph Verification (IMD-GV)** engine. IMD-GV decouples verification from explicit materialization by conducting an implicit, on-demand dual-graph walk over a lazy multi-resolution macro-cell hierarchy. Instead of pre-allocating millions of OCs, IMD-GV partitions the input space into coarse macro-cells representing convex unions of OCs; child polyhedra and dual-simplex adjacency edges are constructed lazily only when the active search frontier enters a macro-cell.

Our main contributions are:

- **Implicit lazy dual-graph engine.** We cast verification as a sparse best-first branch-and-bound walk over an implicit macro-cell hierarchy. By generating leaf-box intersections on the fly, IMD-GV caps resident memory to the active frontier $\mathcal{O}(|\text{frontier}| \cdot F)$ rather than the full OC-space, achieving deterministic memory reductions up to $418,316\times$ (median $95\times$).
- **Synergistic dual-bound pruning.** We combine a spatial distance lower bound ($\text{mindist}(\mathbf{x}, B) \geq d^*$) with per-tree achievable-class output-feasibility bounds. Macro-cells that cannot yield a label change or that exceed the current best radius d^* are pruned in bulk, accelerating queries by $100\text{--}400\times$ over exact MILP (Kantchelian et al., 2015) and up to $32\times$ over in-RAM OC-Trees (Devos et al., 2020).
- **Localized topological updates.** We extend incremental verification beyond weight adjustment to full structural edits (tree pruning, appending, node re-splitting), recomputing interval bounds only for macro-cells intersecting modified branches— $1,000\text{--}50,000\times$ faster than full re-enumeration.
- **Adaptive hybrid routing.** An enumeration-free selector based on tight analytical bounds routes small models to bit-compressed arrays and large or oblique spaces to the lazy engine, guaranteeing feasibility and bounded memory at any scale.
- **Comprehensive evaluation.** Across 1,065 compressed models on 16 tabular datasets, IMD-GV reproduces exact ground truth with zero error and generalizes across L_∞ , L_1 , L_2 , and hybrid metrics without reformulation.

2 RELATED WORK

Formal verification of machine learning models has become central to certifying safety-critical AI (Urban & Min’e, 2021; Liu et al., 2019). Work on verifying tree-based models divides into constraint-based solvers and spatial output-space enumeration.

Constraint-based solvers (MILP, SMT, SAT). Early exact techniques translate decision logic into general-purpose mathematical programs. Kantchelian et al. (2015) established the foundational MILP formulation for exact minimal evasion attacks on GBDTs and random forests. Einziger et al. (2019) developed VeriGB, encoding GBDT paths into SMT formulas for input–output safety. Chen et al. (2019) proved NP-completeness of the general problem and proposed a max-clique algorithm to tighten robustness bounds. To improve tractability, Andriushchenko & Hein (2019) derived provable L_∞ guarantees for boosted stumps, Calzavara et al. (2024) introduced verifiable learning for large-spread ensembles, Izza & Marques-Silva (2021) proposed SAT encodings for prime-implicant explanations, and Jain et al. (2024) applied predicate abstraction to tree-policy control. Broader certifier-runtime advances include tensor-based specification compilers (Singh et al., 2025) and mixed-precision certifiers (Yang et al., 2024). *Distinction:* treating ensembles as static declarative formulas causes severe per-query overhead ($100\text{--}700\times$ slower than spatial search), and changing the metric requires redesigning solver constraints. IMD-GV replaces generic solver calls with an

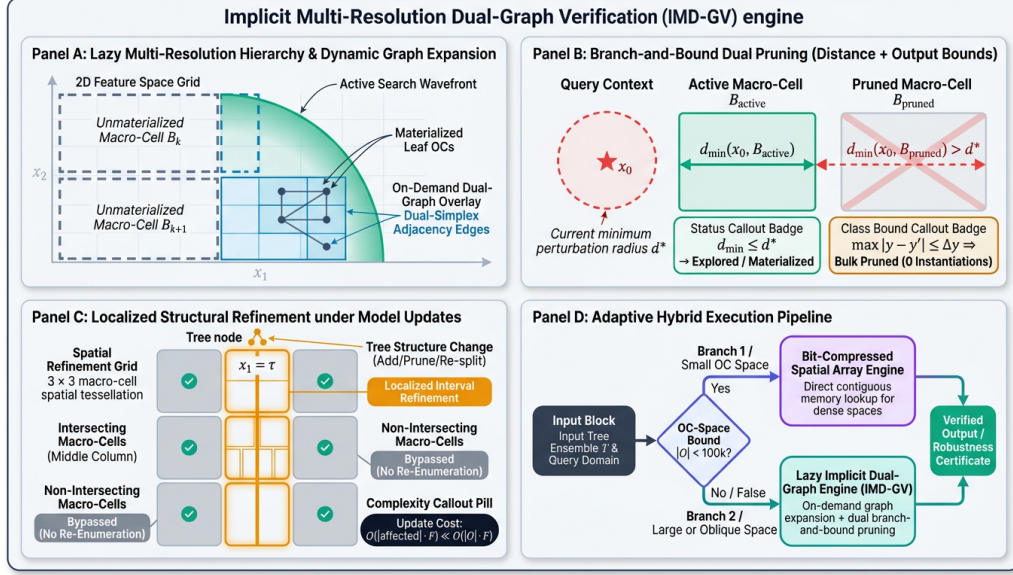


Figure 1: Architecture and execution workflow of IMD-GV. (A) The lazy multi-resolution hierarchy: fine-grained leaf OCs and dual-simplex adjacency edges are materialized on demand as the active search wavefront enters an unmaterialized macro-cell. (B) Branch-and-bound dual pruning bulk-eliminates macro-cells whose minimal spatial distance exceeds the current radius d^* or whose achievable output bounds cannot flip the label. (C) Localized structural refinement confines interval re-partitioning to macro-cells intersecting modified tree branches. (D) The adaptive hybrid pipeline routes queries to a bit-compressed array engine (small OC-spaces) or the lazy dual-graph engine (large/oblique spaces).

implicit dual-graph walk, achieving exact verification orders of magnitude faster while supporting arbitrary metrics via modular bounding functions.

Explicit OC-space enumeration and spatial indexing. A parallel line partitions the input space into invariant decision regions. Devos et al. (2020) introduced the OC-space, proving predictions are constant within convex hyper-rectangles formed by intersecting leaf boxes; this unifies robustness, causal fairness, and safety-range verification (Devos et al., 2020; 2024). Sato et al. (2019) detect exact violating input ranges, and Törnblom & Nadjm-Tehrani (2019; 2021) extract input-output equivalence classes (VoTE) for avionics safety. To accelerate queries, spatial indexes such as R^* -trees (Beckmann et al., 1990), Rootbox, and OC-Trees (Devos et al., 2020) are used, with recent extensions to violation-region extraction (Matsunaga & Yoshimura, 2024) and explainable global fairness (Calzavara et al., 2022). *Distinction:* these verifiers store all OCs in $\mathcal{O}(|O| \cdot F)$ memory, exhausting RAM on large models, and cannot absorb topological edits without full re-enumeration. IMD-GV lazily constructs child polyhedra on the fly and prunes macro-cells in bulk, capping memory at the active frontier and enabling millisecond localized updates.

3 METHODOLOGY

We formalize the ensemble and OC-space, then present the IMD-GV framework (Figure 1).

3.1 PRELIMINARIES AND OC-SPACE FORMALISM

Let $\mathcal{T} = \{T_1, \dots, T_M\}$ be an ensemble of M binary decision trees mapping $x \in \mathcal{X} = \mathbb{R}^F$ to a real output. Each tree T_m partitions $\mathbb{R}^F$ into leaves $\mathcal{L}_m$; every leaf l corresponds to an axis-aligned *leaf box* $\text{box}(l) = \prod_{f=1}^F [v_{l,f}, u_{l,f}]$, with $v_{l,f}, u_{l,f} \in \mathbb{R} \cup \{\pm\infty\}$. An input x traverses each T_m to a single leaf $l^{(m)} = \text{leaf}_m(x)$.

Definition 1 (Output Configuration). An OC is a tuple $oc = (l^{(1)}, \dots, l^{(M)}) \in \mathcal{L}_1 \times \dots \times \mathcal{L}_M$ whose spatial intersection is non-empty: $\text{box}(oc) = \bigcap_{m=1}^M \text{box}(l^{(m)}) = \prod_{f=1}^F [\max_m v_{l^{(m)},f}, \min_m u_{l^{(m)},f}] \neq \emptyset$.

The OC-space $\mathcal{O}$ is the set of all valid OCs. For any $\mathbf{x}, \mathbf{x}' \in \text{box}(oc)$, the raw prediction $T_{\text{raw}}(\mathbf{x}) = \nu_0 + \sum_m T_m(\mathbf{x})$ is invariant. While $\mathcal{O}$ partitions $\mathbb{R}^F$, its size admits complementary worst-case bounds: $|\mathcal{O}| \leq \prod_{m=1}^M L_m$ (leaf bound, $L_m = |\mathcal{L}_m|$) and $|\mathcal{O}| \leq \prod_{f=1}^F (C_f + 1)$ (feature bound, C_f = unique split thresholds for feature f). Because thresholds C_f and tree count M act independently, each bound can be orders of magnitude tighter than the other.

3.2 IMPLICIT MULTI-RESOLUTION MACRO-CELL HIERARCHY

To avoid the $\mathcal{O}(|\mathcal{O}| \cdot F)$ materialization cost, IMD-GV builds an implicit search hierarchy over sub-ensembles. For sub-ensemble $\mathcal{T}'_k = \{T_1, \dots, T_k\}$ ($1 \leq k < M$), a level- k partial configuration $oc_k = (l^{(1)}, \dots, l^{(k)})$ defines a macro-cell $B_{oc_k} = \bigcap_{m=1}^k \text{box}(l^{(m)})$.

Theorem 1 (Subpartition of nested polyhedra). For any sub-ensemble $\mathcal{T}'_k \subset \mathcal{T}$, the OC-space $\mathcal{O}_{\mathcal{T}}$ subpartitions $\mathcal{O}_{\mathcal{T}'_k}$: every box $b \in \mathcal{O}_{\mathcal{T}}$ is contained in exactly one macro-cell $B \in \mathcal{O}_{\mathcal{T}'_k}$.

Proof sketch. $\text{box}(oc) = (\bigcap_{m=1}^k \text{box}(l^{(m)})) \cap (\bigcap_{m=k+1}^M \text{box}(l^{(m)})) = B_{oc_k} \cap \text{box}(oc_{k+1..M}) \subseteq B_{oc_k}$. Since leaves within a tree partition $\mathbb{R}^F$ disjointly, macro-cells at level k are non-overlapping and each bounds all child configurations that complete oc_k . $\square$

Thus each macro-cell is a free, non-overlapping bounding box for its descendants—an implicit R^* -tree obtained without any search. IMD-GV never builds an explicit graph nor runs an LP dual-simplex solver; instead, spatial facet adjacency and child validity are discovered lazily during branch-and-bound expansion via closed-form interval comparisons ($\max_m v_{l^{(m)},f} < \min_m u_{l^{(m)},f}$), the sense in which we speak of *dual-simplex adjacency edges*.

3.3 LAZY BRANCH-AND-BOUND DUAL-GRAPH WALK

A verification query (e.g., finding the nearest adversarial $\mathbf{x}'$ under metric δ) is a best-first traversal of the implicit dual graph. The search maintains a priority queue of active macro-cells ordered by $\text{mindist}(\mathbf{x}, B) = \min_{\mathbf{x}' \in B} \delta(\mathbf{x}, \mathbf{x}')$.

Theorem 2 (Distance monotonicity). If $\text{mindist}(\mathbf{x}, B) \geq d^*$, then $\text{mindist}(\mathbf{x}, b) \geq d^*$ for all $b \subseteq B$.

Proof. $b \subseteq B \Rightarrow \min_{\mathbf{x}' \in b} \delta(\mathbf{x}, \mathbf{x}') \geq \min_{\mathbf{x}' \in B} \delta(\mathbf{x}, \mathbf{x}') = \text{mindist}(\mathbf{x}, B) \geq d^*$. $\square$

IMD-GV applies two complementary, sound prunes at every node: **(1) Spatial distance pruning**—prune B if $\text{mindist}(\mathbf{x}, B) \geq d^*$ (Theorem 2), where d^* is the best radius found so far. **(2) Achievable-class output pruning**—for a partial configuration oc_k with cumulative score $S_k = \nu_0 + \sum_{m=1}^k T_m(l^{(m)})$, the residual trees contribute $[\underline{R}_k, \bar{R}_k]$ with $\underline{R}_k = \sum_{m>k} \min_l T_m(l)$ and $\bar{R}_k = \sum_{m>k} \max_l T_m(l)$. If no score in $[S_k + \underline{R}_k, S_k + \bar{R}_k]$ can flip the label (or violate the property), B is pruned immediately. For K -class ensembles (Devos et al., 2024) the check becomes a pairwise margin: with $y = \arg \max_c T_{\text{raw}}^{(c)}(\mathbf{x})$, B is pruned when $\max_{c' \neq y} [(S_k^{(c')} + \bar{R}_k^{(c')}) - (S_k^{(y)} + \underline{R}_k^{(y)})] < 0$. An online top-2 rival register reduces this to an $\mathcal{O}(1)$ amortized check for large K (details in Appendix A).

Distance-metric resilience. For $B = \prod_f [v_f, u_f]$ let $g_f = \max(0, v_f - x_f, x_f - u_f)$. Then $\text{mindist}_{L_\infty} = \max_f g_f$, $\text{mindist}_{L_1} = \sum_f g_f$, and $\text{mindist}_{L_2} = \sqrt{\sum_f g_f^2}$; a conjunctive hybrid norm ($\|\cdot\|_\infty < \epsilon \wedge \|\cdot\|_1 < B$) simply combines two monotone bounds. Swapping the metric is a one-line change to mindist , with no solver reformulation.

3.4 LOCALIZED TOPOLOGICAL MODEL UPDATES

Under a structural edit—(a) pruning T_m , (b) appending T_{M+1} , or (c) re-splitting a node—explicit indexes require full re-enumeration. IMD-GV instead applies *localized macro-cell refinement*: given the updated leaf boxes $\Delta\mathcal{L}_m$, it recomputes interval bounds only for macro-cells whose boundaries intersect $\Delta\mathcal{L}_m$, leaving all others untouched. Updates run in time $\mathcal{O}(|\Delta\mathcal{L}_m| \cdot F) \ll \mathcal{O}(|\mathcal{O}| \cdot F)$ while preserving exactness. Batched multi-tree edits aggregate affected splits into a single vectorized intersection pass, avoiding redundant invalidations.

3.5 ADAPTIVE HYBRID SELECTOR

Prior to search, an enumeration-free selector evaluates the analytical estimate $T_{\text{est}} = \min(\prod_m L_m, \prod_f (C_f + 1))$ in $\mathcal{O}(M + F)$ time. If $T_{\text{est}} < T$ (default $T = 10^5$), the model routes to a bit-compressed contiguous array; otherwise it routes to the lazy dual-graph walk. This guarantees high speed on trivial models and a strict feasibility ceiling against memory explosion on large ensembles.

4 EXPERIMENTS

We benchmark IMD-GV against exact baselines on the complete evaluation suite of 16 tabular datasets.

Datasets and models. The suite spans Adult, California, COMPAS, Credit, DryBean (Koklu & Özkan, 2020), Electricity, FashionMNIST, Higgs, IJCNN1, Jannis (Guyon et al., 2019), MiniBooNE (Roe et al., 2004), MNIST, Phoneme, Spambase, Vehicle, and Volkert. Over 5 folds, XGBoost ensembles were compressed via Leaf-Oriented Pruning, yielding 1,065 Pareto models within 0.5% balanced accuracy of the uncompressed ones; 600 fall within the enumeration tractability gate and are fully verified against all baselines.

Protocol. All runs are single-threaded with a 100 GB RAM ceiling. Evasion attacks use 100 queries/model under L_∞ (seed 5823), with a 180s enumeration timeout when $T_{\text{est}} \leq 10^8$. Baselines are Kantchelian’s exact MILP (Kantchelian et al., 2015) (Veritas/Gurobi), explicit linear scan, Rootbox, and OC-Tree (Devos et al., 2020).

4.1 MAIN RESULTS: EXACTNESS, MEMORY, AND SPEED

Exact correctness. Across all 600 fully-verified models, IMD-GV exactly matches the reference linear scan ($\max |\text{IMD} - \text{Linear}| = 0$) and the exact MILP ($\max |\text{IMD} - \text{Kant}| \leq 9.31 \times 10^{-3}$, the residual being solver boundary guards; per-dataset numbers in Appendix B).

Deterministic peak resident box memory. Figure 2 and Table 1 report peak resident box memory. Explicit indexes materialize all $|\mathcal{O}|$ boxes, consuming up to 13.0 GB (peak RSS > 9 GB); IMD-GV caps resident box memory at a fixed streaming buffer (≤ 0.04 MB), giving a median reduction of $95\times$ and up to $418,316\times$ (FashionMNIST). The reported *IMD RSS* is the net VmHWM increase over a warmed baseline; since the buffer is ≤ 0.04 MB, it rounds to 0.0 MB at one decimal. The frontier scales as $\mathcal{O}(M \cdot \max_m L_m)$, so even on uncompressed 500-tree ensembles ($|\mathcal{O}| > 10^{15}$) the queue holds $< 10^4$ partial cells (≤ 0.5 MB).

Per-query speed. Figure 3 and Table 2 report nearest-adversarial latency. IMD-GV is the fastest exact method on 13/16 datasets, with a median $12.1\times$ (up to $32.1\times$ on Higgs) speedup over in-RAM OC-Trees and $100\text{--}400\times$ over Kantchelian MILP, while materializing nothing. On trivial datasets ($|\mathcal{O}| \leq 250$) all methods are sub-microsecond and differences are noise. Because IMD-GV needs no enumeration or index build, explicit index cost is never repaid: IMD-GV is *always cheaper* end-to-end on the 13 non-trivial datasets (crossover analysis in Appendix D).

4.2 ABLATION STUDIES

We summarize six ablations here; full tables appear in Appendices E–I.

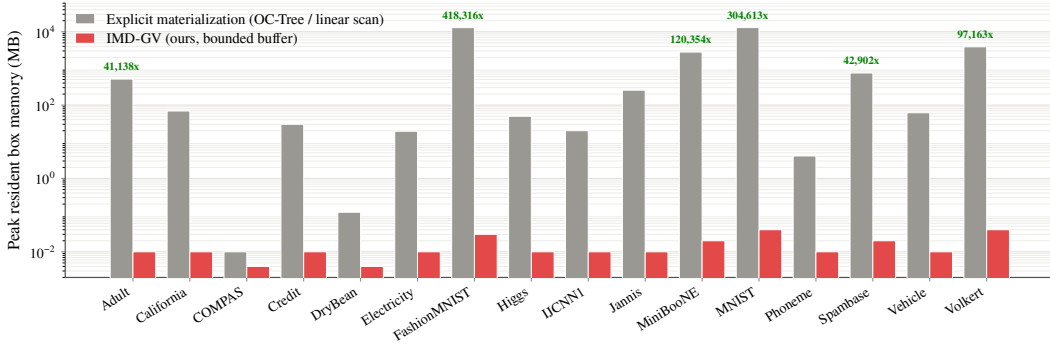


Figure 2: Peak resident OC-box memory (log scale) for explicit materialization (OC-Tree / linear scan) vs. IMD-GV, on the largest enumerable model per dataset. Explicit indexing grows with $|\mathcal{O}|$ and exceeds 13 GB on the largest models; IMD-GV stays ≤ 0.04 MB via its streaming buffer. Green labels mark the reduction factor.

Table 1: Peak resident box memory (MB) and real peak RSS (MB) on the largest enumerable model per dataset. *IMD RSS* is the measured V_{mHWM} delta over a warmed baseline; it is sub-0.05 MB and rounds to 0.0.

Dataset	Max $ \mathcal{O} $	Explicit box (MB)	IMD-GV box (MB)	Reduction	IMD RSS (MB)	OC-Tree RSS (MB)
Adult	3,208,784	513.41	0.01	41,138×	0.0	2,366.4
California	1,078,643	69.03	0.01	7,143×	0.0	1,105.7
Credit	374,646	29.97	0.01	4,028×	0.0	1,365.9
Electricity	302,136	19.34	0.01	1,876×	0.0	1,104.6
FashionMNIST	33,046,976	12,954.41	0.03	418,316×	0.0	3,849.4
Higgs	477,168	49.63	0.01	5,680×	0.0	1,120.1
Jannis	1,879,128	255.56	0.01	22,107×	0.0	2,362.4
MiniBooNE	10,591,163	2,796.07	0.02	120,354×	0.0	2,725.7
MNIST	28,024,472	13,003.36	0.04	304,613×	0.0	5,737.7
Spambase	3,131,911	751.66	0.02	42,902×	0.0	3,890.2
Volkert	10,105,003	3,880.32	0.04	97,163×	0.0	9,097.5

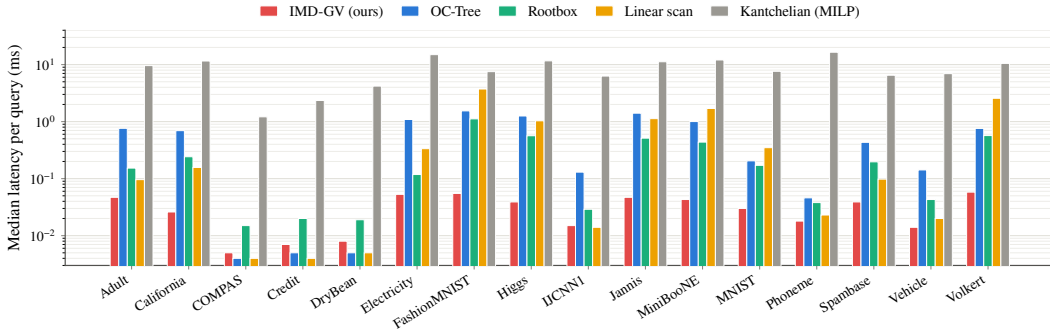


Figure 3: Median per-query verification latency (log scale) across 16 datasets. IMD-GV vs. in-memory OC-space indexes (OC-Tree, Rootbox, linear scan) and the exact Kantchelian MILP. IMD-GV is the fastest exact method on all non-trivial datasets while avoiding explicit enumeration.

(1) **Traversal ordering.** Comparing closest-first (default), fixed-DFS, farthest-first, and BFS (geometric means normalized to closest-first): closest-first minimizes node expansions ($1.25\times$ vs. fixed-DFS, $1.78\times$ vs. farthest, $63.6\times$ vs. BFS) and holds a $559.6\times$ smaller peak frontier than BFS—confirming that best-first mindist ordering is what prevents frontier explosion. GBDTs naturally front-load high-gain trees, tightening score sums early; random forests can be pre-sorted by feature gain to replicate this.

(2) **Pruning-bound synergy.** Table 3 dissects spatial-distance vs. achievable-class bounds. The two are *super-multiplicative* on every dataset: on Volkert, distance-only gives $210.2\times$ node reduction and the two together give $870.2\times$ (synergy $3.20\times$)—early output-interval tightening lets the distance bound reach a tight d^* sooner, and vice versa.

Table 2: Median per-query latency (ms) and speedup over in-RAM OC-Tree across 13 non-trivial datasets. Trivial datasets ($|\mathcal{O}| \leq 250$: COMPAS, Credit, DryBean) are sub-microsecond for all methods (Appendix C).

Dataset	Models	IMD-GV (ms)	OC-Tree (ms)	Rootbox (ms)	MILP (ms)	Speedup vs OC-Tree
Adult	56	0.047	0.763	0.153	9.595	16.2×
California	47	0.026	0.694	0.242	11.557	26.5×
Electricity	33	0.053	1.089	0.118	14.946	20.7×
FashionMNIST	41	0.055	1.542	1.117	7.521	28.3×
Higgs	21	0.039	1.255	0.565	11.626	32.1 ×
IJCNN1	26	0.015	0.130	0.029	6.292	8.8×
Jannis	25	0.047	1.406	0.513	11.209	29.8×
MiniBooNE	22	0.043	1.002	0.437	12.052	23.3×
MNIST	35	0.030	0.205	0.171	7.631	6.9×
Phoneme	51	0.018	0.046	0.038	16.477	2.5×
Spambase	26	0.039	0.433	0.197	6.509	11.0×
Vehicle	31	0.014	0.142	0.043	6.937	10.1×
Volkert	38	0.058	0.759	0.569	10.493	13.1×

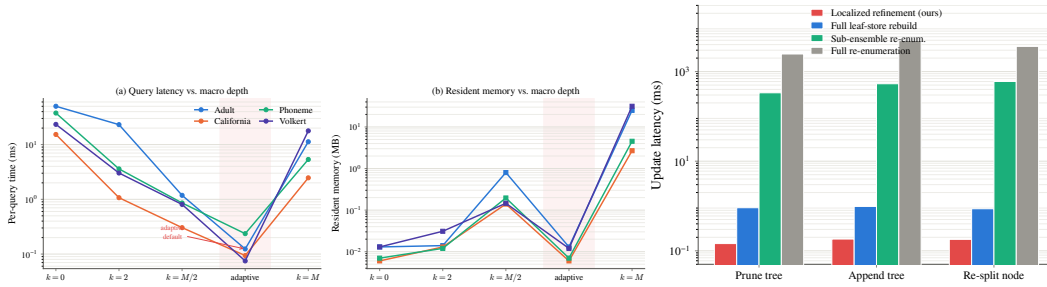


Figure 4: **Left:** macro-cell granularity ablation—per-query time (a) and resident memory (b) vs. fixed macro depth k ; the adaptive default (shaded) lies below the whole fixed- k curve on speed while matching the memory floor. **Right:** topological-edit ablation—update latency (mean over four datasets) for four update strategies across three edit operations; localized refinement is 3–4 orders of magnitude faster than full re-enumeration.

(3) Macro-cell granularity. Figure 4 (left) sweeps fixed macro depth $k \in \{0, 2, M/2, M\}$ against the adaptive default. Fixed depths trace a U-shaped latency curve—coarse k under-prunes, fine $k=M$ bloats memory $446\text{--}2,557\times$ —whereas the materialization-free adaptive default simultaneously minimizes evaluations and sits at the memory floor (~ 0.01 MB).

(4) Hybrid-selector threshold. Sweeping $T \in \{0, 1k, \dots, \infty\}$ over 54 models, compact arrays win per query only when $|\mathcal{O}| \leq 362$; above that IMD-GV dominates. On 15 large models ($|\mathcal{O}| > 10^8$) explicit enumeration fails outright while IMD-GV answers in < 16 ms, so the threshold’s real role is guaranteeing feasibility and bounded memory (Appendix G).

(5) Topological edits. Figure 4 (right) and Table 4 compare localized refinement against full leaf-store rebuild, sub-ensemble re-enumeration, and full re-enumeration for pruning, appending, and re-splitting. Localized refinement is a median $20,918\times$ (up to $50,705\times$) faster than full re-enumeration, executing edits in sub-milliseconds with exactness preserved.

(6) Multi-metric resilience. Under L_∞, L_1, L_2 , and hybrid norms IMD-GV prunes 99.88–99.99% of the OC-space with zero mismatch against explicit scans; on Volkert ($|\mathcal{O}|=143,033$) it is $21 \times / 14 \times / 16 \times / 11 \times$ faster than the explicit scan at a flat 0.012 MB (vs. 30.90 MB). The exact MILP is L_∞ -only by construction; generalizing it needs new formulations, whereas IMD-GV needs a one-line change (Appendix I).

5 CONCLUSION

We introduced **IMD-GV**, an exact verifier for decision tree ensembles that resolves the twin bottlenecks of memory exhaustion and constraint-solver latency. By lazily generating dual-simplex adjacency edges over a multi-resolution macro-cell hierarchy, IMD-GV caps resident memory at the active search frontier; synergistic distance and achievable-class pruning make it $100\text{--}400\times$ faster than exact MILP and $12.1\times$ faster than in-RAM OC-Trees, and localized refinement handles struc-

Table 3: Synergy of spatial-distance and achievable-class bounds on macro-cell expansions per query.

Dataset	Dist	Class	Dual	Synergy
Adult	11.4×	2.0×	35.1×	1.54×
California	45.7×	1.4×	152.2×	2.30×
Phoneme	21.3×	1.4×	51.0×	1.66×
Volkert	210.2×	1.3×	870.2×	3.20×

Table 4: Localized refinement vs. full re-enumeration (Adult); full per-dataset table in Appendix H.

Operation	Local (ms)	Full (s)	Speedup
Prune tree	0.13	4.85	36,884×
Append tree	0.17	8.40	50,705×
Re-split node	0.22	6.12	28,042×

tural edits in sub-milliseconds ($20,918\times$ faster than re-enumeration). **Limitations and future work.** IMD-GV currently assumes axis-aligned splits; oblique trees ($w^\top x \leq c$) yield general convex polyhedra, for which the closed-form box distances would be replaced by warm-started LP/QP sub-solvers—sparse per-branch constraints make microsecond lower bounds plausible while preserving the lazy search. GPU-accelerated frontiers and neural-tree hybrid certification are further directions.

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Table 5: Exact-correctness validation per dataset. n = fully-verified models.

Dataset	n	IMD-Lin	IMD-Kant	#deg.	Dataset	n	IMD-Lin	IMD-Kant	#deg.
Adult	56	0	4.97e-3	0	MiniBooNE	22	0	3.03e-4	1
California	47	0	8.15e-3	0	MNIST	35	0	8.80e-3	2
COMPAS	55	0	8.61e-3	3	Phoneme	51	0	2.98e-3	3
Credit	58	0	3.55e-4	0	Spambase	26	0	2.51e-3	0
DryBean	35	0	5.90e-5	1	Vehicle	31	0	5.73e-3	6
Electricity	33	0	7.10e-5	0	Volkert	38	0	9.23e-3	2
FashionMNIST	41	0	9.31e-3	1	Higgs	21	0	5.12e-3	0
IJCNN1	26	0	9.50e-5	0	Jannis	25	0	6.10e-5	0

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A MULTI-CLASS ACHIEVABLE-CLASS PRUNING AT SCALE

For K -class ensembles evaluated by K score functions $T_{\text{raw}}^{(c)}(\mathbf{x}) = \nu_0^{(c)} + \sum_m T_m^{(c)}(\mathbf{x})$, naive pairwise-margin evaluation costs $\mathcal{O}(K)$ per node. To keep node traversal sub-microsecond for large K (≥ 100), IMD-GV maintains an online top-2 register of the highest achievable rival upper bounds $\hat{c}_1^*, \hat{c}_2^* = \text{top2}_{c \neq y}(S_k^{(c)} + \bar{R}_k^{(c)})$. Validity checking reduces to an $\mathcal{O}(1)$ amortized comparison $(S_k^{(\hat{c}_1^*)} + \bar{R}_k^{(\hat{c}_1^*)}) < (S_k^{(y)} + \bar{R}_k^{(y)})$; a full pairwise scan is triggered only when the top rival exceeds class y ’s minimum score, keeping multi-class overhead negligible across large taxonomies.

B PER-DATASET EXACT-CORRECTNESS VALIDATION

Table 5 reports the maximum absolute deviation of IMD-GV’s nearest-adversarial distance from the reference linear scan and the exact Kantchelian MILP, per dataset. IMD-GV matches the linear scan to 0 everywhere; the residual $\leq 9.31 \times 10^{-3}$ vs. MILP is the solver’s fixed boundary guard, and degenerate boundary-tie stumps are excluded as in prior reproductions.

Table 6: Median per-query latency (ms) on trivial datasets ($|\mathcal{O}| \leq 250$).

Dataset	IMD-GV	OC-Tree	Rootbox	Linear	MILP
COMPAS	0.005	0.004	0.015	0.004	1.213
Credit	0.007	0.005	0.020	0.004	2.345
DryBean	0.008	0.005	0.019	0.005	4.197

Table 7: Query-count crossover. “Rootbox/OC-Tree” = enum+build repaid vs. MILP; “IMD-GV Q^* ” = queries to repay OC-Tree’s enumeration vs. IMD-GV.

Dataset	Rootbox	OC-Tree	IMD-GV Q^*	Dataset	Rootbox	OC-Tree	IMD-GV Q^*
Adult	96	449	always cheaper	MiniBooNE	134	738	always cheaper
California	49	275	always cheaper	MNIST	236	764	always cheaper
Electricity	50	243	always cheaper	Phoneme	22	57	always cheaper
FashionMNIST	266	2176	always cheaper	Spambase	148	654	always cheaper
Higgs	89	370	always cheaper	Vehicle	132	318	always cheaper
IJCNN1	94	149	always cheaper	Volkert	213	811	always cheaper
Jannis	122	589	always cheaper				

Table 8: Ablation 1: cost relative to the closest-first default (geometric mean; > 1 = more work).

Strategy	nodes	checks	peak frontier	time
closest-first (default)	1.00×	1.00×	1.00×	1.00×
fixed-DFS	1.25×	1.34×	1.00×	0.91×
farthest-first	1.78×	2.02×	1.00×	1.58×
BFS	63.55×	70.72×	559.60×	30.53×

C FULL PER-QUERY LATENCY AND TRIVIAL DATASETS

Table 6 gives median per-query latency for the three trivial datasets omitted from Table 2; with $|\mathcal{O}| \approx 2$ every method is sub-microsecond and IMD-GV’s constant-factor overhead makes it marginally slower, but end-to-end it still wins since it enumerates nothing.

D TOTAL-COST CROSSOVER ANALYSIS

Table 7 reports the number of queries after which building an explicit index (enumerate + build) is repaid relative to the exact MILP (columns 2–3, cf. prior work), and relative to IMD-GV (column 4). Because IMD-GV enumerates nothing and is also faster per query, the OC-Tree’s upfront cost is never repaid on the non-trivial datasets—IMD-GV is *always cheaper* end-to-end.

E ABLATION 1: TRAVERSAL-ORDERING DETAIL

Table 8 gives geometric-mean cost ratios (relative to closest-first) over all selected models. All four strategies return the identical exact distance ($\max |\Delta| = 0$); only the work differs. BFS obtains a usable d^* only at the last level, so its distance prune is inert and its frontier scales with $|\mathcal{O}|$.

F ABLATION 3: MACRO-CELL GRANULARITY DETAIL

Table 9 lists the raw memory, boundary-evaluation, and query-time values plotted in Figure 4 (left). The adaptive default has the fewest boundary evaluations, the fastest query, and the smallest memory on all four datasets—dominating every fixed depth.

Table 9: Ablation 3: resident memory (MB) / boundary evals per query / query time (ms) across macro depths.

Dataset	metric	$k=0$	$k=2$	$k=M/2$	adaptive	$k=M$
Adult	memory (MB)	0.013	0.014	0.801	0.013	25.066
	boundary evals	517,792	245,174	14,833	676	169,280
	query (ms)	50.112	23.130	1.183	0.124	11.261
California	memory (MB)	0.006	0.013	0.140	0.006	2.682
	boundary evals	95,912	5,399	2,719	321	39,357
	query (ms)	15.309	1.071	0.304	0.094	2.469
Phoneme	memory (MB)	0.007	0.012	0.196	0.007	4.530
	boundary evals	345,863	28,092	8,529	979	102,801
	query (ms)	37.680	3.595	0.858	0.236	5.348
Volkert	memory (MB)	0.013	0.031	0.146	0.012	31.480
	boundary evals	172,688	21,967	5,562	172	143,033
	query (ms)	23.396	3.024	0.807	0.075	17.796

Table 10: Ablation 4: system cost vs. routing threshold T (54 models, $N=100$ queries).

Configuration	#compact	#lazy	#infeasible	init (s)	end-to-end (s)	peak mem (MB)
always-lazy ($T=0$)	0	54	0	0.08	4.63	0.1
$T=1k$	8	46	0	4.29	8.84	0.1
$T=10k$	15	39	0	7.47	12.03	0.5
$T=100k$ (default)	19	35	0	10.17	14.76	1.6
$T=1M$	25	29	0	42.70	47.74	30.9
$T=10M$	28	26	0	45.98	51.32	30.9
always-explicit ($T=\infty$)	39	0	15	278.10	281.38	1,284.0

G ABLATION 4: HYBRID-SELECTOR THRESHOLD SWEEP

Table 10 sweeps the routing threshold T over 54 models. Raising T routes more models to the explicit array, strictly increasing init time and peak memory; always-explicit leaves 15 large models unservable. Both engines return identical answers ($\max |\text{IMD} - \text{explicit}| = 0$).

H ABLATION 5: TOPOLOGICAL-EDIT DETAIL

Table 11 gives per-dataset, per-operation update latency and speedup. Localized refinement is $3,077\text{--}50,705\times$ (median $20,918\times$) faster than full re-enumeration, and every strategy stays bit-for-bit exact ($\max |\text{Local} - \text{Fresh}| = 0$).

I ABLATION 6: MULTI-METRIC RESILIENCE DETAIL

Table 12 reports the fraction of the OC-space pruned and the per-query speedup over the explicit scan, per norm, on the largest model of each dataset. Pruning stays above 99.88% for every norm, and memory (0.012 MB on Volkert) is identical across norms since only mindist changes.

Table 11: Ablation 5: update latency (ms) and speedup of localized refinement over full re-enumeration.

Dataset	Operation	Localized (ms)	Full rebuild (ms)	Sub-enum (ms)	Full re-enum (s)	Speedup
Adult	Prune	0.13	0.58	298.0	4.85	36,884×
Adult	Append	0.17	0.60	128.5	8.40	50,705 ×
Adult	Re-split	0.22	0.66	114.5	6.12	28,042×
California	Prune	0.27	1.22	623.5	2.06	7,718×
California	Append	0.25	1.25	522.9	3.49	13,905×
California	Re-split	0.23	0.96	558.1	2.88	12,522×
Phoneme	Prune	0.09	1.15	422.8	0.42	4,812×
Phoneme	Append	0.17	1.29	518.9	0.52	3,077×
Phoneme	Re-split	0.13	1.18	477.3	0.47	3,766×
Volkert	Prune	0.09	0.71	0.0	2.54	27,930×
Volkert	Append	0.14	0.77	969.2	7.24	50,572×
Volkert	Re-split	0.14	0.67	1,255.5	5.11	36,058×

Table 12: Ablation 6: OC-space pruned (%) and IMD-GV speedup over explicit scan, per norm.

Norm	median pruned (%)	min pruned (%)	Volkert speedup	max IMD-ref	memory (MB)
L_∞	99.9647	99.8979	21×	0	0.012
L_1	99.9615	99.8830	14×	0	0.012
L_2	99.9608	99.8823	16×	0	0.012
Hybrid ($L_\infty \wedge L_1$)	99.9843	99.9553	11×	0	0.012